

Problem Set: Numerical Methods — Key

Geophysics 3A

1. Governing equation

The steady-state heat conduction equation with spatially variable thermal conductivity and internal heat production is:

$$\nabla \cdot (k \nabla T) + A = 0 \quad (1)$$

In two dimensions (x, z) :

$$\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + A = 0 \quad (2)$$

2. Boundary conditions

(a) Top and bottom

Surface:

$$T(x, 0) = T_s \quad (3)$$

Basal boundary:

$$T(x, z_b(x)) = T_b(x) \quad (4)$$

(b) Side boundaries (insulating)

Zero heat flux:

$$\frac{\partial T}{\partial x} = 0 \quad (5)$$

Finite difference form:

$$T_{i,1} = T_{i,2}, \quad T_{i,N} = T_{i,N-1} \quad (6)$$

3. Finite difference equation

Using fluxes across cell faces:

$$\frac{1}{\Delta x^2} \left[k_{i+\frac{1}{2},j} (T_{i+1,j} - T_{i,j}) - k_{i-\frac{1}{2},j} (T_{i,j} - T_{i-1,j}) \right] \quad (7)$$

$$+ \frac{1}{\Delta z^2} \left[k_{i,j+\frac{1}{2}} (T_{i,j+1} - T_{i,j}) - k_{i,j-\frac{1}{2}} (T_{i,j} - T_{i,j-1}) \right] + A_{i,j} = 0 \quad (8)$$

Rearranged:

$$T_{i,j} = \frac{k_{i+\frac{1}{2},j} T_{i+1,j} / \Delta x^2 + k_{i-\frac{1}{2},j} T_{i-1,j} / \Delta x^2 + k_{i,j+\frac{1}{2}} T_{i,j+1} / \Delta z^2 + k_{i,j-\frac{1}{2}} T_{i,j-1} / \Delta z^2 + A_{i,j}}{k_{i+\frac{1}{2},j} / \Delta x^2 + k_{i-\frac{1}{2},j} / \Delta x^2 + k_{i,j+\frac{1}{2}} / \Delta z^2 + k_{i,j-\frac{1}{2}} / \Delta z^2} \quad (9)$$

4. Heat production at nodes

(a) General expression:

$$A_{i,j} = \frac{1}{4} \left(A_{i-\frac{1}{2},j-\frac{1}{2}} + A_{i+\frac{1}{2},j-\frac{1}{2}} + A_{i-\frac{1}{2},j+\frac{1}{2}} + A_{i+\frac{1}{2},j+\frac{1}{2}} \right) \quad (10)$$

(b) If $A = A(z)$ only:

All four surrounding cells have values depending only on depth:

$$A_{i,j} = \frac{1}{2} (A_{i-\frac{1}{2}} + A_{i+\frac{1}{2}}) \quad (11)$$

If the node depth coincides with the midpoint:

$$A_{i,j} \approx A(z_i) \quad (12)$$

5. Thermal conductivity in four directions

Conductivities at interfaces:

$$k_{i+\frac{1}{2},j} = \frac{1}{2} \left(k_{i+\frac{1}{2},j-\frac{1}{2}} + k_{i+\frac{1}{2},j+\frac{1}{2}} \right) \quad (13)$$

$$k_{i-\frac{1}{2},j} = \frac{1}{2} \left(k_{i-\frac{1}{2},j-\frac{1}{2}} + k_{i-\frac{1}{2},j+\frac{1}{2}} \right) \quad (14)$$

$$k_{i,j+\frac{1}{2}} = \frac{1}{2} \left(k_{i-\frac{1}{2},j+\frac{1}{2}} + k_{i+\frac{1}{2},j+\frac{1}{2}} \right) \quad (15)$$

$$k_{i,j-\frac{1}{2}} = \frac{1}{2} \left(k_{i-\frac{1}{2},j-\frac{1}{2}} + k_{i+\frac{1}{2},j-\frac{1}{2}} \right) \quad (16)$$

6. Porosity-controlled properties

(a) Thermal conductivity:

$$k = \left(\frac{1-\phi}{k_m} + \frac{\phi}{k_w} \right)^{-1} \quad (17)$$

(b) Heat production:

$$A = A_0(1 - \phi) \quad (18)$$

(c) **Qualitative behaviour**

- Porosity decreases with depth
- Conductivity increases with depth (less fluid, more solid matrix)
- Heat production increases with depth (more radioactive solid material)

7. Surface heat flow

Heat flow given by Fourier's law:

$$q = -k \frac{\partial T}{\partial z} \quad (19)$$

Finite difference approximation:

$$q_j \approx -k_{1,j} \frac{T_{2,j} - T_{1,j}}{\Delta z} \quad (20)$$

8. Conceptual question

Although k and A vary only with depth, the geometry of the system (sediment thickness) varies laterally.

This creates lateral temperature gradients, which cause heat to flow horizontally as well as vertically.

Therefore, the system is two-dimensional because:

- boundary geometry varies in x
- isotherms deform around basement topography
- heat is redistributed laterally

Key point: Geometry, not material properties, drives the 2-D behaviour.

9. Sketching exercise

1-D case:

- Flat sediment layer
- Horizontal isotherms
- Vertical heat flow

2-D case:

- Variable sediment thickness
- Isotherms bend upward over highs
- Heat flow diverges over highs and converges in basins

10. Model comparison

- 1-D model provides smooth, averaged behaviour
- 2-D model captures local variations due to bathymetry
- 2-D model fits short-wavelength variability better
- Both models may miss some systematic trends

11. 3-D effects

- 2-D ridge: heat deflection occurs in one horizontal direction
- Seamount (3-D): heat spreads radially
- 3-D spreading reduces peak anomalies

Including 3-D effects can improve the model but will not fully remove misfit.

12. Improving the model

Possible improvements:

- Include hydrothermal circulation
- Allow lateral variation in material properties
- Improve boundary conditions (e.g. basal temperature structure)
- Use fully 3-D geometry